THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	$72 = 2^3 \times 3^2$; multiply by the primes with odd power
2	**(b)** 3^5	Add the exponents on top, then subtract the bottom exponent
3	**(a)** 22	Expand with powers of the base
4	**(d)** 75	$4x + 60 = 360$
5	**(d)** 2	Alternate digit sums must differ by 0 or 11
6	**(c)** $6x^2 + 19x + 10$	Multiply term by term and collect like terms
7	**(a)** 63	One part $= 9$
8	**(a)** 1200	CP = SP×100/(100+5)
9	**(c)** 24	Height splits the base: $h^2 = 30^2 - 18^2 = 576$
10	**(a)** 2:3	A:B:C $= 6 : 8 : 9$
11	**(d)** 8	$n = 6$, faces $= n + 2$
12	**(c)** 52	$(10 \times 40 + 15 \times 60) \div 25$
13	**(c)** $5x + 12$	$3x + 12 + 2x$
14	**(d)** 252 cm²	$300 - 48$
15	**(b)** 3	$192 = 2^6 \times 3$
16	**(a)** $\frac{9}{4}$	$(a/b)^{-n} = (b/a)^n$
17	**(b)** 101101	Repeated division by 2
18	**(b)** 20 cm	$2k(3 + 5) = 64$ gives $k = 4$
19	**(d)** 0	The number must be even and its digit sum a multiple of 3
20	**(a)** $4x^2 - 12xy + 9y^2$	$(a - b)^2 = a^2 - 2ab + b^2$
21	**(a)** 24	Subtract the added number
22	**(d)** 1800	CP = SP×100/(100−15)
23	**(b)** 84	Other side $= 24$; perimeter $= 2(18 + 24)$
24	**(b)** 24	Total work $= 12 \times 8 \times 15 = 1440$ man-hours
25	**(b)** 12	Base has $n = 6$ sides; edges $= 2n$
26	**(c)** 42	Sum changes by 6
27	**(a)** $x^2 + 7x + 10$	Multiply term by term
28	**(a)** 12	$b_1h_1 = b_2h_2$: $180/15$
29	**(d)** 9	Only the ones digit 7 matters: 7^2 ends in 9
30	**(b)** 7	$x + 4 = 11$
31	**(a)** CXI	44+67=111
32	**(b)** 34 cm	AO = OC; find x first, then AC = 2 AO
33	**(c)** 8262	Largest arrangement minus smallest arrangement
34	**(a)** 2600	$(63 + 37)(63 - 37) = 100 \times 26$
35	**(b)** 18 L	$\frac{3}{10} \times 60$
36	**(c)** 1800	MP = SP×100/(100−15)
37	**(b)** 48	Half of other diagonal $= \sqrt{30^2 - 18^2} = 24$
38	**(a)** 720	One part $= 1350 \div 15 = 90$
39	**(d)** 12	Each edge is shared by 2 faces: $E = 3F/2$
40	**(c)** 42.5	Arrange in ascending order first
41	**(d)** 15	$3 \times 9 - 5 \times 4 + 8$
42	**(d)** 216 cm²	Other half-diagonal $= \sqrt{15^2 - 9^2} = 12$; area $= \frac{1}{2} \times 18 \times 24$

43	**(c)** 1.2	1.44 is the square of 1.2
44	**(b)** 4	$3x = 12$
45	**(c)** 7	It is $111_2 = 2^3 - 1$
46	**(d)** 17	Adjacent angles are supplementary
47	**(c)** 60	Find the LCM
48	**(c)** 100	$(7 + 3)^2$
49	**(d)** 15 hours	More pipes, less time
50	**(b)** 3000	P = SI×100/(R×T)

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 30	Height difference 18 m; $\sqrt{18^2+24^2}=30$
2	**(a)** 30	Quantity × price is constant
3	**(a)** 20	$V=2+E-F$
4	**(d)** 40	Required total 180 minus 140
5	**(c)** $10x+14$	Perimeter $=2(l+b)$
6	**(d)** 156 m²	$(20+4)(15+4)-20\times15$
7	**(c)** 0.3	$0.027=0.3^3$
8	**(b)** 1000	$a^m\times b^m=(ab)^m=(10)^3$
9	**(d)** 5	20 lies between 2^4 and 2^5
10	**(c)** 85°	$\angle B=\angle D$ and the angle sum is 360°
11	**(d)** 3	4829 leaves remainder 5 when divided by 8
12	**(c)** 54	$(4+2)(4+5)$
13	**(a)** 55	A : B : C = 4 : 6 : 5, so one part is 11
14	**(d)** 240	Original = result × 100/125
15	**(d)** 216	Other leg $=24$; area $=\frac{1}{2}ab$
16	**(c)** 12	Distance $=540$ km
17	**(d)** 8	Inner cube of side 2
18	**(a)** 20	$k(n+1)/2=4\times10/2$
19	**(d)** 53	$(x+y)^2=x^2+y^2+2xy$
20	**(c)** 150 cm²	Other leg $=\sqrt{25^2-15^2}=20$; area $=\frac{1}{2}\times15\times20$
21	**(c)** 2025	Numbers ending in 5: 4×5 then 25 gives 2025
22	**(a)** 6×10^7	Multiply the numbers, add the exponents, then adjust
23	**(c)** 10001_2	11+6=17
24	**(d)** 85°	Exterior angles of a quadrilateral add up to 360°
25	**(a)** 1001	Smallest 4-digit number is 1000; add what is needed to reach a multiple of 7
26	**(c)** $8x$	The x^2 and constant terms cancel
27	**(d)** 2 km	1 km = 100000 cm
28	**(d)** 8	Difference = P(R/100)² = 5000×(4/100)²
29	**(b)** 33, 44, 55	Multiples of $(3,4,5)$ by 11 stay a triplet
30	**(b)** 32	One part $=8$; difference $=(11-7)\times8$
31	**(d)** 48	Length multiplies by $\frac{4}{3}$ each stage: $27\times(4/3)^2$
32	**(d)** 50%	Rise $=10$; $(10/20)imes100$
33	**(d)** $6n+6$	$2n+(2n+2)+\ldots:6n+6$
34	**(b)** 50 cm²	Area $=d^2/2=10^2/2$
35	**(a)** 2	$98=2\times7^2$; multiply by the primes with odd power
36	**(a)** 431	$512-81=431$
37	**(c)** 100	Digit x (base)^(position from right, starting 0)
38	**(b)** rectangle	A parallelogram with equal diagonals
39	**(d)** 45	Subtract the reverse
40	**(c)** $2x+22$	Expand each bracket and collect like terms
41	**(b)** 10 m/s	Multiply by $\frac{5}{18}$
42	**(d)** 1620	2000×90/100×90/100

43	**(b)** 36	Height splits the base: $h^2 = 39^2 - 15^2 = 1296$
44	**(b)** 32	$80/360 \times 144$
45	**(a)** 13	$1 + 3 + \cdots + 3^2 = (3^3 - 1)/2$
46	**(a)** 41	$(20 \times 35 + 30 \times 45) \div 50$
47	**(c)** $9x + 10$	$5x + 10 + 4x$
48	**(b)** 380 cm²	$450 - 70$
49	**(b)** 2	$250 = 2 \times 5^3$
50	**(c)** $\frac{5}{6}$	$\frac{1}{2} + \frac{1}{3}$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 27	Expand with powers of the base
2	**(c)** 60	$4x + 120 = 360$
3	**(c)** 2	Alternate digit sums must differ by 0 or 11
4	**(c)** $12x^2 + 11x + 2$	Multiply term by term and collect like terms
5	**(a)** 48	One part $= 6$
6	**(a)** 2000	CP = SP×100/(100+20)
7	**(d)** 102	Other side $= 36$; perimeter $= 2(15 + 36)$
8	**(d)** 4:5	A:B:C $= 4 : 14 : 5$
9	**(b)** 10	$n = 8$, faces $= n + 2$
10	**(d)** 36	Sum changes by 12
11	**(d)** $x^2 + 7x + 12$	Multiply term by term
12	**(a)** 15	$b_1h_1 = b_2h_2$: $240/16$
13	**(d)** 4	Only the ones digit 8 matters: 8^2 ends in 4
14	**(b)** 5^4	Add the exponents on top, then subtract the bottom exponent
15	**(d)** 110110	Repeated division by 2
16	**(b)** 18 cm	$2k(2 + 3) = 60$ gives $k = 6$
17	**(b)** 0	The number must be even and its digit sum a multiple of 3
18	**(a)** $9x^2 - 6xy + y^2$	$(a - b)^2 = a^2 - 2ab + b^2$
19	**(a)** 12	Subtract the added number
20	**(a)** 3500	CP = SP×100/(100−20)
21	**(d)** 72	Half of other diagonal $= \sqrt{39^2 - 15^2} = 36$
22	**(b)** 18	Total work $= 9 \times 6 \times 20 = 1080$ man-hours
23	**(a)** 16	Base has $n = 8$ sides; edges $= 2n$
24	**(a)** 40.5	Arrange in ascending order first
25	**(c)** 40	$4 \times 12 - 6 \times 3 + 10$
26	**(a)** 480 cm²	Other half-diagonal $= \sqrt{26^2 - 10^2} = 24$; area $= \frac{1}{2} \times 20 \times 48$
27	**(d)** 1.5	2.25 is the square of 1.5
28	**(a)** $\frac{25}{9}$	$(a/b)^{-n} = (b/a)^n$
29	**(a)** CCXII	78+134=212
30	**(c)** 20 cm	AO = OC; find x first, then AC = 2 AO
31	**(b)** 6084	Largest arrangement minus smallest arrangement
32	**(a)** 5600	$(78 + 22)(78 - 22) = 100 \times 56$
33	**(d)** 24 L	$\frac{2}{7} \times 84$
34	**(d)** 2800	MP = SP×100/(100−20)
35	**(c)** 39	Height difference 15 m; $\sqrt{15^2 + 36^2} = 39$
36	**(a)** 480	One part $= 780 \div 13 = 60$
37	**(c)** 18	Each edge is shared by 2 faces: $E = 3F/2$
38	**(b)** 0	Required total 150 minus 150
39	**(a)** $14x + 14$	Perimeter $= 2(l + b)$
40	**(b)** 324 m²	$(30 + 6)(18 + 6) - 30 \times 18$
41	**(c)** 0.5	$0.125 = 0.5^3$
42	**(c)** 7	$x + 2 = 9$

43	**(b)** 26	It is $222_3 = 3^3 - 1$
44	**(a)** 14	Adjacent angles are supplementary
45	**(a)** 24	Find the LCM
46	**(d)** 100	$(6+4)^2$
47	**(c)** 6 hours	More pipes, less time
48	**(d)** 4500	P = SI×100/(R×T)
49	**(d)** 270	Other leg $= 36$; area $= \frac{1}{2}ab$
50	**(b)** 24	Quantity × price is constant

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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